Pooled cloglog GLMM: talk prep

Briefing for today's meeting with the postdoc · 11 November 2026 · condensed from *reports/Pooled cloglog GLMM evaluation.md*

The answer in one breath

Keep the segment-level cloglog GLMM as the primary model; no non-ML alternative is clearly better. But change the argument for it: the random intercept does **not** make it predict better. It earns its place because it (1) gives honest standard errors, so the 5-df likelihood-ratio test of the geometry block is valid, and (2) is a joint model of a person's segments, which is what lets nine segment risks be combined into one person risk. The weak spot is that person risk: it rests on σ^2 and a normal frailty shape that nine binary segments per person identify poorly.

The model in four lines

- One row per coronary segment (about nine per person); outcome: new plaque between the two scans.
- cloglog link, so coefficients are hazard ratios and segment hazards add up to a person hazard.
- Log segment length as exposure; random person intercept $b_i \sim N(0, \sigma^2)$ (a shared frailty); segment position as fixed factor; adaptive Gauss–Hermite quadrature.
- Person risk = $1 - E_b[\exp(-S_i(b))]$, integrated over b, never with b = 0.

What the evidence says (three findings to remember)

- **No prediction gain.** Integrated GLMM predictions tie with GEE and plain logistic regression (calibration slopes 0.97–0.98, Pavlou 2015); setting b = 0 is *worse* (slope 0.83 vs 0.97, Bouwmeester 2013). So drop any "predicts better" sentence.
- **At or above coronary practice on clustering.** Only 5 of 15 intracoronary imaging studies handled clustering (Lingsma 2013). Cohort papers use GEE or marginal Cox; shear-stress studies use mixed models.
- **Unusual on three points.** No coronary paper read uses a cloglog link, a length offset, or pooling segment risks to a patient risk. Defend them from the survival argument, not from convention: with two scans, the model *is* the discrete-time form of a shared-frailty Cox model.

Weaknesses, ranked

Weakness	How serious	What answers it
σ^2 weakly identified; person risk depends on $\hat{\sigma}^2$ and the normal shape	High, for person-level calibration	Fold-wise $\hat{\sigma}$, gamma-frailty refit, person-level calibration and benchmark
Segments not independent given the person (left-main features on adjacent pLAD/pLCx rows)	Moderate–high for those features	Co-occurrence check, GEE robust Wald test, person-cluster bootstrap
Fixed offset on log length (hazard proportional to length)	Moderate; can leak into curvature and taper	Free coefficient (1 df) as default
Riley criteria assume independent observations	Moderate	Count persons, not rows; estimate shrinkage by resampling
LR test is asymptotic	Low–moderate with ~1,000 persons	Parametric bootstrap of the LR statistic
cloglog vs logit	Low for the test; matters for the person risk	Logit refit as sensitivity

Alternatives considered

Alternative	Verdict
Person-level cloglog on per-person geometry summaries	Equal for “who”; add as **benchmark**
Working-independence cloglog GEE + robust 5-df Wald	Worse as primary (no person risk); best **check on the LR test**
Interval-censored / discrete-time shared-frailty survival	The same model with two scans; use as **justification**
Bayesian GLMM, weakly informative priors	Equal at best; optional
Vessel-within-person intercept	Optional sensitivity; likely a boundary estimate
Conditional logistic / between-within terms	Worse for prediction; useful confounding check (+2 df)
Ridge, lasso, Firth GLMM	Worse: unreliable at small n, loses the LR test
Beta-binomial; random slopes; spatial effects	Worse or not estimable with nine rows

Decisions to get the postdoc's view on

- **Log length: free coefficient or fixed offset?** Recommend free coefficient in the base model (interval reported against 1), offset as sensitivity.
- **Left-main features:** keep them on the pLAD and pLCx rows, or test them at person level? Recommend deciding before the test is locked, with a person-level test of the three features beside the block test.
- **Primary test:** single 5-df geometry block (current notes) versus the between-person block from the earlier report. This report keeps the 5-df block; between-within terms move to sensitivity.
- **Event rate:** the notes assumed rare events, but an earlier report gives ~452 of 1,076 persons with new plaque (~42%, not checked against current data). Tabulate the segment-level rate and the number of persons with two or more positive segments before locking the model.
- **How many sensitivity analyses** to prespecify (list below), and whether a small simulation at the fitted parameters is worth the time.

Likely questions, with short answers

Why not GEE, like the CCTA cohort papers?

GEE gives valid marginal segment risks but no joint model, so it cannot give the probability that *any* of a person's segments develops plaque; the independence product overstates it. GEE stays as the robust-Wald check on the LR test.

Why cloglog and not logit?

At segment level the fit can't tell them apart. The reason is interpretive: hazard ratios that don't depend on window length, and segment hazards that add to a person hazard. Logit refit as sensitivity.

Does the random intercept improve prediction?

No. For a new person, integrated predictions tie with a model ignoring clustering. It is there for valid inference and for aggregation.

Are your hazard ratios conditional or marginal?

Conditional on the person (the frailty). Person-level predictions integrate over b, so they are marginal.

Is σ^2 estimable with nine segments per person?

Only weakly: it is informed mostly by persons with two or more positive segments. That is why fold-wise $\hat{\sigma}$, a gamma-frailty refit and person-level calibration are planned.

Your sample size: persons or segments?

Persons. Riley's criteria assume independent observations, and three of the five geometry features are person-level constants, so segment rows would overstate the budget.

Why not penalise the geometry block?

Penalisation is least reliable at small effective n (Riley 2021) and removes the chi-square reference for the confirmatory test.

What does TRIPOD-Cluster say?

It explicitly excludes clustering within individuals, so it applies by analogy: say how clustering was handled, which intercept is used for predictions, and keep every step inside the resampling.

Sensitivity analyses to propose (in order of value)

1. Person-level cloglog model with one prespecified summary per feature (benchmark; a “who” test).
2. Working-independence cloglog GEE, robust 5-df Wald test, plus person-cluster bootstrap.
3. Parametric bootstrap of the LR statistic.
4. Gamma-frailty refit (closed-form person risk).
5. Fold-wise $\hat{o}$ from several starting values; report boundary estimates.
6. Logit refit; fixed-offset length.
7. Person means of curvature and taper (+2 df, between-within).
8. Number of evaluable segments as covariate; calibration by segment count.
9. Observed versus model-implied plaque co-occurrence in adjacent and non-adjacent segment pairs.
10. Vessel-within-person intercept.

Be careful what you quote today

Numbers from Pavlou, Bouwmeester, Falconieri, Moineddin, Austin and Riley were relayed by a summarising tool; check them against the PDFs before they go in the thesis. Pavlou's “slope 0.79 vs 0.98” is a logit result at ICC 0.30 for individual predictions: use it for the direction of the b = 0 error, not its size here. PROSPECT, PROSPECT II, EMERALD, PREDICTION, TRIPOD+AI and PROBAST could not be read (publisher blocks). No published simulation covers ~1,000 clusters of nine, so claims about σ^2 bias here are reasoned, not sourced.